Foreign Investors as Leaders or Followers? Evidence from Foreign Portfolio Investment in India

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Abstract

Foreign Portfolio Investment (FPI) plays a central role in emerging financial markets. In India, movements of FPI receive significant energy from financial media, investors, and policymakers, where FPI is associated with driving returns in Indian equity markets and holding predictive power for future market direction. This view assumes foreign investors hold superior information compared to domestic institutional and retail investors, under this perception, FPI inflows should precede positive market returns, while FPI outflows should precede market declines. This paper investigates whether Indian equity returns attract subsequent FPI inflows using monthly data from March 2021 to March 2026. Ordinary Least Squares (OLS) regression models are estimated using lagged NIFTY 50 returns, the CBOE Volatility Index (VIX), and lagged FPI flows.

The results indicate that lagged market returns remain positively associated with subsequent FPI inflows across multiple specifications, while higher levels of global risk are associated with lower foreign investment. These findings are consistent with positive feedback trading, suggesting that foreign investors may respond to recent market performance rather than systematically anticipate future returns. However, the possibility that both equity returns and foreign investment flows respond to common information or broader macroeconomic conditions cannot be excluded. Therefore, the results should be interpreted as evidence consistent with foreign investors behaving more as followers than leaders of Indian equity-market movements, rather than as conclusive evidence of a causal feedback-trading mechanism.

I. Introduction

Financial markets perform an important informational function in addition to allocating capital. Hayek (1945) argued that economic knowledge is dispersed among individuals and that the price system communicates information that cannot be fully possessed by any single participant. In financial markets, this provides a useful theoretical basis for considering whether observed price movements contain information that investors subsequently incorporate into their decisions.

This idea is closely related to the Efficient Market Hypothesis (EMH), under which asset prices incorporate available information. If foreign portfolio investors possess superior information or process information more effectively than other market participants, their investment decisions could contribute to price discovery. Under this interpretation, FPI flows should precede subsequent equity returns, as foreign investors act on information before it is fully reflected in market prices.

However, the relationship may operate in the opposite direction. Foreign investors may respond to information already reflected in market prices rather than anticipate future movements. Positive feedback trading provides one possible explanation, whereby investors increase their exposure following strong market performance and reduce exposure following weak performance. Under this interpretation, previous equity returns should be associated with subsequent FPI flows.

A previous study by the author examined this hypothesis using monthly Indian data from March 2021 to March 2026. Although contemporaneous FPI flows were significantly associated with NIFTY 50 returns, lagged FPI flows failed to predict subsequent market performance. These findings provided little support for the informed trading hypothesis and suggested that foreign investors may not systematically anticipate future returns at monthly frequencies.

The first study examines whether foreign investors lead Indian equity markets. This study examines the reverse possibility: whether foreign investors respond to market movements that have already occurred.

This raises an alternative explanation. Rather than leading markets, foreign investors may allocate capital after observing strong market performance. Such behaviour is consistent with positive feedback trading, in which investors increase exposure following periods of strong returns and reduce exposure following weaker market conditions.

This paper investigates whether Indian equity returns attract subsequent foreign portfolio investment inflows. Specifically, it asks whether previous NIFTY 50 returns are associated with future FPI activity after accounting for global risk sentiment and persistence in investment flows.

The paper contributes to the existing analysis in two ways. First, it examines a competing explanation for the relationship between foreign investment and equity returns identified in earlier work. Second, it shifts the focus from whether foreign investors predict returns to whether they respond to them.

The findings suggest that lagged market returns remain positively associated with subsequent FPI inflows across multiple specifications. However, these results should not be interpreted as conclusive evidence of feedback trading, since both returns and investment flows may respond to common information shocks or broader macroeconomic developments. The evidence therefore suggests that foreign investors may behave more as followers than leaders of market movements, while acknowledging that alternative explanations remain plausible.

II. Literature Review and Theoretical Framework

2.1 Information, Prices, and Market Efficiency

Hayek (1945) argued that economically relevant information is dispersed among individuals and that the price system communicates information that no single participant possesses in its entirety. In financial markets, this provides a theoretical basis for considering how information incorporated into market prices may subsequently influence investors' decisions.

This idea is closely related to the Efficient Market Hypothesis (EMH), developed by Fama (1970). Under the semi-strong form of market efficiency, security prices incorporate publicly available information, limiting the ability of investors to systematically earn excess returns using information that is already publicly available. Applied to foreign portfolio investment, an informed-investor interpretation would imply that investors possessing relevant information trade before that information is fully reflected in market prices. Under this interpretation, FPI flows could precede subsequent movements in equity returns.

However, market efficiency does not imply that investors disregard past market performance. Previous price movements may contain information about changing expectations, economic conditions, or investor sentiment, and investors may adjust their portfolios in response. Therefore, a relationship between previous equity returns and subsequent FPI flows would not, by itself, demonstrate market inefficiency or irrational investor behaviour.

This distinction is important for the present study. The objective is not to test whether the Indian equity market satisfies the EMH, but to examine the directional relationship between market returns and subsequent foreign investment flows.

2.2 Foreign Investors, Information, and Price Discovery

Foreign investors are important participants in emerging financial markets and may contribute to price discovery through their trading decisions. One possible explanation is that foreign investors possess information that is not yet fully incorporated into domestic prices. If this mechanism dominates, foreign investment activity should precede subsequent equity-market returns.

Empirical evidence from emerging markets provides some support for this informed-investor perspective. Cai, Ho, and Zhang (2022), examining 1,504 stocks across 23 emerging markets, find a reliable relationship between direct foreign ownership and intraday price discovery. Their analysis further indicates that private information plays a substantially greater role than investor recognition or unexplained factors in facilitating price discovery. These findings suggest that foreign investors can play an information-relevant role in the price-formation process.

A previous study by the author examined this possibility in the Indian context using monthly data from March 2021 to March 2026. The study found a statistically significant contemporaneous association between FPI flows and NIFTY 50 returns. However, when FPI flows were lagged by one month, their relationship with subsequent NIFTY 50 returns became statistically insignificant. Thus, the results provided little evidence that previous-month FPI flows systematically predict subsequent NIFTY 50 returns at the monthly frequency.

This finding does not establish that foreign investors lack superior information. Rather, it weakens the evidence for the specific directional proposition that previous FPI activity reliably precedes subsequent Indian equity returns. This motivates examination of the reverse direction: whether market performance precedes subsequent foreign investment flows.

2.3 Positive Feedback Trading

An alternative explanation is positive feedback trading, in which investors adjust their positions in response to previous market performance. Under this framework, investors increase their exposure following relatively strong returns and reduce their exposure following weaker returns. The resulting relationship runs from previous market returns to subsequent investment flows.

Evidence from other emerging markets supports the relevance of this mechanism. Krohn, Sushko, and Synsatayakul (2023) examine foreign-investor trading behaviour across the foreign-exchange, equity, and fixed-income markets of Thailand. They find that foreign investors engage in momentum trading, reinforcing positive feedback between the direction of trading and returns. They also find that local financial investors tend to follow foreign investors with a lag, while foreign order-flow innovations contain information about future returns.

The findings of Krohn et al. (2023) are particularly relevant because they demonstrate that feedback trading and information-based trading need not be mutually exclusive. Foreign investors may respond to previous returns while also possessing information relevant to subsequent price movements. Consequently, observing a positive relationship between previous returns and subsequent FPI flows would be consistent with positive feedback trading, but would not establish that foreign investors are uninformed or irrational.

For the present study, the central empirical implication is that a positive and statistically significant coefficient on lagged NIFTY 50 returns would indicate that stronger recent market performance is associated with greater subsequent FPI inflows.

2.4 Common Information and Alternative Explanations

A third possibility is that neither FPI flows nor equity returns directly causes the other. Instead, both variables may respond to common information or underlying economic conditions. Monetary-policy expectations, economic growth expectations, geopolitical developments, corporate earnings expectations, and global investor sentiment could influence both Indian equity prices and foreign investment decisions.

This possibility is particularly important when interpreting a positive relationship between lagged NIFTY 50 returns and subsequent FPI flows. A strong market return may itself incorporate information about improving economic or corporate prospects, which foreign investors subsequently observe and respond to. In such a case, the observed relationship would reflect information transmission through prices rather than a purely mechanical tendency to chase returns.

The CBOE Volatility Index (VIX) is therefore included as a control for global risk sentiment. Higher VIX levels generally indicate greater uncertainty in global financial markets and may reduce investors' willingness to allocate capital to emerging-market equities. However, VIX cannot capture every factor affecting either Indian equity returns or FPI decisions. Its inclusion therefore reduces, but does not eliminate, the possibility of omitted-variable bias or other sources of endogeneity.

2.5 Research Gap and Hypothesis Development

The literature therefore provides three competing interpretations of the relationship between foreign investment and equity-market performance. The informed-investor perspective suggests that foreign investment activity may precede subsequent returns, while positive feedback trading suggests that previous market performance may instead precede foreign investment flows. A third possibility is that both variables respond to common information, meaning that an observed statistical relationship does not necessarily represent a direct causal mechanism.

The previous study examined the first direction and found that lagged FPI flows did not significantly predict subsequent NIFTY 50 returns. The present study therefore examines the reverse direction: whether previous NIFTY 50 returns are associated with subsequent FPI inflows, while controlling for global risk sentiment and persistence in FPI flows.

III. Research Question

The paper investigates the following question:

Do Indian equity returns attract subsequent Foreign Portfolio Investment inflows?

This leads to two hypotheses:

H1: Informed Trading Hypothesis

Foreign investors possess superior information and anticipate future market movements.

$$FPI_{t-1} \rightarrow Return_t$$

Prediction:

- Lagged FPI predicts future returns.

This hypothesis was not supported in the previous study.

H2: Positive Feedback Trading Hypothesis

Foreign investors increase allocations after observing strong market performance.

$$Return_{t-1} \rightarrow FPI_t$$

Prediction:

- Lagged returns predict future FPI inflows.

IV. Methodology

This study examines whether recent stock market performance is associated with subsequent Foreign Portfolio Investment (FPI) inflows into the Indian equity market. Monthly data covering the period from March 2021 to March 2026 are used, providing sixty observations after accounting for lagged variables. The analysis focuses on net FPI equity flows as the dependent variable and employs Ordinary Least Squares (OLS) regression to estimate the relationship between lagged market returns and subsequent foreign investment activity. Monthly observations are chosen to maintain consistency with the previous study and to capture medium-term investment behaviour rather than short-term trading decisions.

The dependent variable is monthly net Foreign Portfolio Investment (FPI) equity inflows measured in Indian Rupees (₹ crore). The primary explanatory variable is the previous month's NIFTY 50 return, measured in percentage terms. This variable tests the central hypothesis that foreign investors allocate capital after observing recent market performance. The CBOE Volatility Index (VIX) is included as a proxy for global investor risk sentiment, as periods of heightened uncertainty may discourage investment in emerging markets irrespective of domestic market performance. Lagged FPI flows are incorporated to account for persistence in institutional investment decisions, recognising that portfolio reallocations may occur gradually rather than instantaneously.

Four regression models are estimated. The first model examines the relationship between lagged NIFTY returns and subsequent FPI inflows without additional controls. The second model introduces the VIX to account for changes in global risk sentiment. The third model further includes lagged FPI flows to control for persistence in investment behaviour. Finally, the fourth model replaces the contemporaneous VIX with its lagged value so that all explanatory variables belong to the previous month's information set.

As a diagnostic for potential multicollinearity, pairwise correlations were calculated among the explanatory variables in the preferred specification. Lagged FPI exhibited relatively weak correlations with lagged NIFTY 50 returns (−0.277) and lagged VIX (−0.177), while lagged NIFTY 50 returns and lagged VIX showed a moderate positive correlation of 0.617. These correlations do not indicate an obvious severe multicollinearity problem, although the moderate relationship between market returns and VIX warrants consideration when interpreting their individual coefficients.

Ordinary Least Squares (OLS) regression is employed because the objective of the study is to estimate the association between recent market performance and subsequent foreign investment while controlling for selected explanatory variables. The analysis is intended to identify statistical relationships rather than establish definitive causality. Consequently, the results should be interpreted as evidence consistent with particular theories of investor behaviour rather than conclusive proof of causal mechanisms.

Model Based Methodology

Model 1: Baseline specification

$$FPI_t = \alpha + \beta_1 R_{t-1} + \varepsilon_t$$

Model 2: Controlling for global risk sentiment

$$FPI_t = \alpha + \beta_1 R_{t-1} + \beta_2 VIX_t + \varepsilon_t$$

Model 3: Controlling for FPI persistence

$$FPI_t = \alpha + \beta_1 R_{t-1} + \beta_2 VIX_t + \beta_3 FPI_{t-1} + \varepsilon_t$$

Model 4: Preferred specification

$$FPI_t = \alpha + \beta_1 R_{t-1} + \beta_2 VIX_{t-1} + \beta_3 FPI_{t-1} + \varepsilon_t$$

Where:

- FPI_t= net Foreign Portfolio Investment equity flows in month t, measured in ₹ crore.
- R_{t-1}= NIFTY 50 percentage return in month $t-1$.
- VIX_t= CBOE Volatility Index in month t.
- VIX_{t-1}= CBOE Volatility Index in month $t-1$.
- FPI_{t-1}= net FPI equity flows in month $t-1$.
- α= regression intercept.
- $\beta_1, \beta_2, \beta_3$= coefficients measuring the conditional association between each explanatory variable and current FPI flows.

- ε_t= error term capturing other factors affecting FPI flows not included in the model.

V. Results

Table 1. Regression Results

Variable	Model 1	Model 2	Model 3	Model 4
Lagged NIFTY Return	335,240***	347,937***	459,058***	347,855**
	(0.013)	(0.007)	(0.003)	(0.031)
VIX	—	−2,300***	−2,568***	—
		(0.010)	(0.005)	
Lagged FPI	—	—	−0.20	−0.139
			(0.192)	(0.371)
Lagged VIX	—	—	—	−1,627*
				(0.095)
Constant	−1,795	43,090	51,773	28,215
R^2	0.101	0.199	0.224	0.151
Adjusted R^2	0.085	0.171	0.182	0.106
n	60	60	60	60

Notes: Values in parentheses are p-values.
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

The results of the four regression models indicate a consistent positive relationship between previous month's NIFTY 50 returns and subsequent Foreign Portfolio Investment inflows. In the baseline specification, lagged market returns are positively associated with FPI inflows and are statistically significant at the 5 percent level. This provides initial evidence that higher previous-month NIFTY 50 returns are positively associated with subsequent FPI inflows.

Introducing the VIX as a control for global risk sentiment does not materially alter the main finding. Lagged market returns remain positive and statistically significant, while the coefficient on the VIX is negative and statistically significant, indicating that higher levels of global uncertainty are associated with lower foreign investment in Indian equities. The improvement in explanatory power suggests that both domestic market performance and global risk conditions influence foreign portfolio allocation decisions.

The third specification includes lagged FPI flows to account for persistence in institutional investment behaviour. Although the coefficient on lagged FPI is statistically insignificant, lagged market returns continue to exhibit a positive and statistically significant relationship with subsequent FPI inflows. This suggests that the observed relationship is not merely the result of gradual execution of investment decisions across multiple months. Instead, previous market performance appears to contain independent explanatory power even after accounting for investment persistence.

The final specification replaces the contemporaneous VIX with its lagged value to ensure that all explanatory variables are determined before the dependent variable. The principal findings remain unchanged. Lagged market returns continue to display a positive and statistically significant association with subsequent FPI inflows, while lagged VIX maintains the expected negative relationship with foreign investment, although with weaker statistical significance. Overall, the consistency of the lagged return coefficient across all four specifications suggests that the results are robust to alternative model specifications.

Robustness Check

Variable	Model 4	Robustness Check
Lagged NIFTY 50 Return	347,855 **	302,008 **
	(0.031)	(0.036)
Lagged VIX	−1,627*	−1,337
	(0.095)	(0.121)
Lagged FPI	−0.139	−0.006
	(0.371)	(0.967)
Constant	28,215	23,226
	(0.132)	(0.162)
R^2	0.151	0.180
Adjusted R^2	0.106	0.135
n	60	58

Notes**:** Coefficients are reported with p-values in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The robustness specification excludes September 2024 and October 2024, the two observations with the largest absolute FPI flows.

As a robustness check, the preferred specification was re-estimated after excluding the two observations with the largest absolute FPI flows. The coefficient on lagged NIFTY 50 returns remained positive and statistically significant, although it declined from approximately ₹348 crore to ₹302 crore per one-percentage-point increase in lagged returns. The p-value increased only slightly, from 0.031 to 0.036. This suggests that the primary relationship is not driven solely by the two most extreme FPI observations.

VI. Discussion

The findings provide evidence that is more consistent with positive feedback trading than with the informed trading hypothesis. The previous study found that lagged FPI flows did not significantly predict future NIFTY 50 returns, providing little support for the view that foreign investors systematically anticipate market movements. In contrast, the present analysis indicates that recent market performance is positively associated with subsequent FPI inflows. Together, these findings suggest that foreign investors may respond to observed market performance rather than consistently leading it.

However, these results should not be interpreted as conclusive evidence of feedback trading. An alternative explanation is that both equity returns and foreign portfolio investment respond to common information shocks, including changes in global economic conditions, monetary policy expectations, geopolitical developments, or broader investor sentiment. Although the inclusion of the VIX partially accounts for global risk conditions, it cannot capture all potential sources of omitted variable bias. Consequently, the regressions identify robust statistical associations rather than definitive causal relationships.

Taken together, the findings contribute to the debate regarding the informational role of foreign investors in emerging equity markets. Rather than supporting the traditional view that foreign investors consistently lead market movements through superior information, the evidence suggests that their investment behaviour may be better characterised as responding to recent market performance. Nevertheless, distinguishing between positive feedback trading and common-information effects requires more sophisticated empirical techniques than those employed in this study.

VII. Limitations

Several limitations should be acknowledged. First, the analysis is based on monthly observations over a five-year period, resulting in a relatively small sample size. Second, Ordinary Least Squares regression identifies statistical relationships but cannot establish causality. Third, although the VIX serves as a useful proxy for global investor sentiment, other macroeconomic and financial variables may simultaneously influence both stock market returns and foreign investment flows. Finally, the use of monthly data may overlook short-term adjustments in foreign portfolio allocation that occur at daily or weekly frequencies.

Future research could extend this analysis by employing higher-frequency data, vector autoregression, Granger causality tests, or instrumental variable techniques to better distinguish between informed trading, positive feedback trading, and common-information effects.

VIII. Conclusion

This study examined whether Indian equity returns attract subsequent Foreign Portfolio Investment inflows using monthly data from March 2021 to March 2026. Across four regression specifications, previous month's NIFTY 50 returns remained positively associated with subsequent FPI inflows, while higher levels of global risk generally reduced foreign investment. In contrast, lagged FPI flows did not exhibit consistent explanatory power once additional controls were introduced.

When considered alongside the findings of the previous study, the evidence suggests that foreign investors do not systematically predict future Indian equity returns. Instead, their behaviour appears consistent with responding to recent market performance. However, the possibility that both market returns and foreign investment flows react to common information cannot be excluded. Accordingly, the findings should be interpreted as evidence consistent with positive feedback trading rather than proof of a causal relationship.

This study contributes to the literature by examining the reverse direction of the relationship between foreign portfolio investment and equity returns in the Indian market. While the results provide support for the hypothesis that foreign investors behave more as followers than leaders of market movements, further research using richer datasets and more advanced econometric methods is required to identify the underlying causal mechanisms more precisely.

References

Cai, J., Ho, R. Y. K., & Zhang, Z. (2022). Foreign investors, private information, and price discovery. *International Review of Economics & Finance, 80*, 506–525.

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance, 25*(2), 383–417.

Hayek, F. A. (1945). The use of knowledge in society. *The American Economic Review, 35*(4), 519–530.

Krohn, I., Sushko, V., & Synsatayakul, W. (2023). Foreign investor feedback trading in an emerging financial market (BIS Working Papers No. 1154). *Bank for International Settlements.*

Thakkar, V. (2026). Do foreign portfolio investors drive Indian equity returns, or do they simply follow market momentum? *SSRN.*